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Ensembling and Random Forests

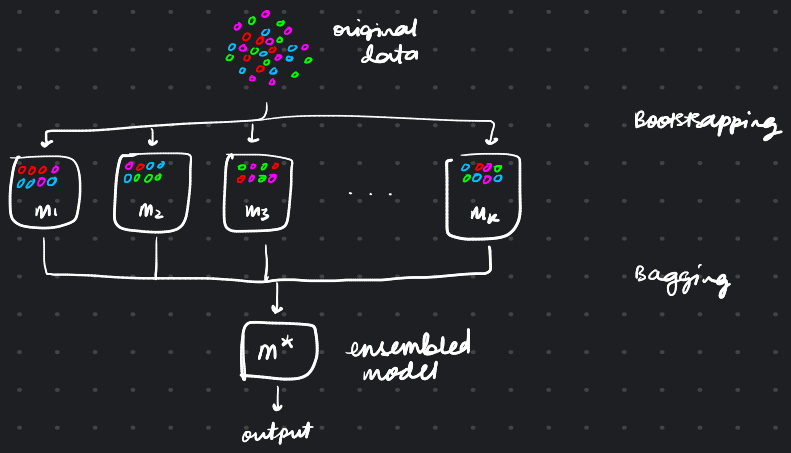
Ensembling techniques are powerful methods in ML that combine multiple ML models to improve predictive performance. By aggregating from various models, ensembling reduces overfitting, improves accuracy, and enhances robustness compared to individual models.

Ensembling
- Bagging (bootstrap aggregating) : multiple instances of the same learning algo are trained on diff. subsets of data.
- Boosting : models are trained sequentially. Each subsequent model focuses on correcting the errors made by the previous models.

Bagging (bootstrap aggregating)

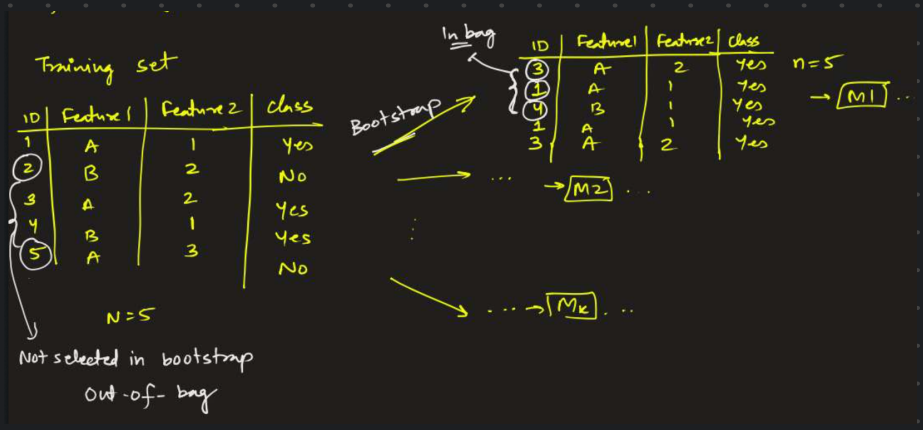
Suppose a set of N examples (D) are given for training, each containing p dimensional input vectors.
At each iteration i, a training set D_i of n tuples is selected via row sampling with replacement method from D (Bootstrap).
Then a classifier model M_i is learned for each training set D_i.
Each classifier returns its class prediction. The bagged classifier M* counts the votes and assigns the class with the most votes to an unknown sample.

Bootstrap sampling: Random selection with replacement.
Random feature selection: A random subset of features is considered for training models.
Aggregation: Majority voting (classification), average the predictions (regression).



Bootstrapping

Bootstrapping is a statistical resampling technique that involves repeatedly sampling with replacement from a dataset.
From the original set of size N, multiple samples of size n (generally n=N) are created. Each point in each sample is drawn from the initial dataset randomly, allowing for the possibility of selecting the same instance multiple times.



$$\text{Out-of-bag sample} = \frac{\#\text{ of data points not in bag in a bootstrap}}{\#\text{ of data points in a dataset}}$$

In bagging, the out-of-bag data points can be used as the validation set for the corresponding model.

$$\text{Bagging fraction} = \frac{\#\text{ of unique data points in the bag}}{\#\text{ of data points in the dataset}}$$

Bagging fraction $\approx 63.2\%$ if $n = N$ and N is large.

Proof:

$$P(\text{rejected}) = \left(1 - \frac{1}{N}\right)^n = \left(1 - \frac{1}{N}\right)^N$$

If N is large,

$$\lim_{N \to \infty} P(\text{rejected}) = \lim_{N \to \infty} \left(1 - \frac{1}{N}\right)^N = \frac{1}{e} \approx 0.3679$$

P that a particular element is NOT selected $\approx 36.79\%$

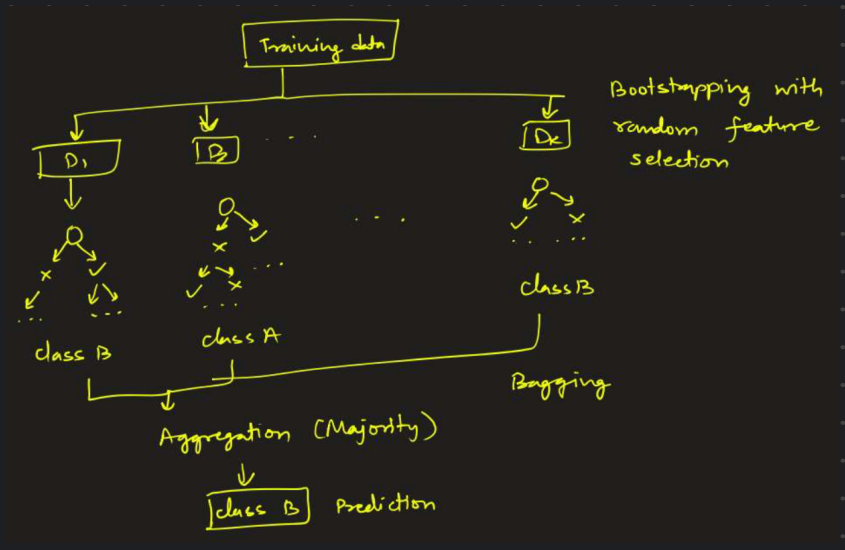
is selected $\approx 63.2\%$

$$\lim_{x \to \infty} f(x)^{g(x)} = e^{\lim_{x \to \infty} (f(x) - 1) \cdot g(x)}$$

$$\lim_{N \to \infty} \left(1 - \frac{1}{N}\right)^N = e^{\lim \left(-\frac{1}{N}\right) \cdot N} = e^{-1}$$

Random forests

It is an extension of bagging that enhances the decision tree algorithm. We build multiple decision trees on the bootstrapped samples with random feature selection and then aggregate the results produced by these decision trees.



Advantages:
- high accuracy
- robustness
- reduced overfitting
- feature importance
- versatile

Disadvantages:
- complexity
- memory consumption
- longer training time
- bins in predictions